

Arsitha Sathu

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SUMMARY

Master's graduate in Computer Science with practical experience building and scaling backend and full-stack applications using Java, Spring Boot, React, Node.js, and cloud services. Strong background in designing APIs, developing microservices, and implementing real-time features, including voicebot integrations. Skilled at debugging complex issues, improving performance, and delivering reliable production-quality code. Seeking software engineering roles where I can deepen my expertise and contribute to high-impact systems.

EDUCATION

George Mason University Masters, Computer Science (GPA: 3.83) **Aug 2023 – May 2025**
Jawaharlal Technological University Bachelors, Computer Science Engineering (GPA: 9.21) **Aug 2017 – June 2021**

SKILLS

Programming & Methodologies: C, Java, Python, SQL, JavaScript, TypeScript, HTML, CSS, JSON, Bash, Shell Scripting, REST APIs, GraphQL, Microservices, Agile, Scrum

Frameworks & Tools: Spring Boot, Node.js, React.js, Angular, JUnit, Jest, Postman, Swagger, SonarQube, VS Code, IntelliJ, Git, Bitbucket

Technical Skills: PostgreSQL, MySQL, MongoDB, Unix, Windows, Data Structures & Algorithms, Operating Systems, OOP Concepts, DBMS, Web Development, Data Mining, NLP, LLMs

DevOps & Automation: Docker, Jenkins, CI/CD Pipelines, AWS (EC2, S3), Azure (AI Services, Container Apps, Key Vault)

Coursework: Data Structures & Algorithms, Operating Systems, OOPS Concepts, DBMS, Web Development, Software Engineering, Data Mining, NLP, LLMs

EXPERIENCE

Blue Horse Digital | *Software Engineer (volunteer)* **Jul 2025 – present**

- Built the core speech-to-speech Voicebot workflow using Azure Container Apps, Azure AI Foundry (GPT-4o Realtime), and Azure Key Vault.
- Developed backend services in FastAPI and integrated Azure Speech SDK for smooth STT/TTS interactions.
- Implemented WebSocket communication to support low-latency, real-time audio exchange with the GPT-4o model.
- Automated CI/CD pipelines to build and deploy containerized services through Azure Container Registry and Azure Container Apps.
- Centralized secret management with Azure Key Vault, strengthening security and ensuring consistent, safe deployments.

Infosys Limited | *Fullstack Developer* **Oct 2021 – Jul 2023**

- Built fullstack features using Java, Spring Boot, Angular, and TypeScript, developing REST APIs, UI components, and database logic for production systems.
- Designed and optimized microservices integrated with PostgreSQL, improving API performance by 20–30% through query tuning and refactoring slow endpoints.
- Created reusable frontend components (dynamic tables, filters, role-based views) that reduced duplicated code and standardized dashboards across teams.
- Debugged and resolved production issues by analyzing logs, reproducing failures, and implementing fixes across backend and frontend layers.
- Maintained high-quality code by applying SonarQube standards and writing JUnit tests, consistently achieving more than 95% coverage.

PROJECTS

AI Log Analysis | *Python, React, JavaScript, Spring Boot, GPT-4o-mini, PostgreSQL, Git*

- Built a full-stack AI log analysis system using Spring Boot, PostgreSQL, FastAPI, and GPT-4o-mini to analyze logs and generate debugging insights.
- Developed a React dashboard for one-click log analysis, visualizing error clusters and recommended fixes across services.

Departmental Connect | *ReactJS, HTML, CSS, Bootstrap, Java Script, Node.js, SQL, Git*

- Redesigned a static departmental website into a dynamic React and Node.js platform with searchable content and a responsive UI, enabling faster navigation and easier content management.

Abstractive text summarization | *Python, mT5 Transformer, Git, PyTorch, Transformers, Wandb, Pandas, NumPy*

- Fine-tuned an mT5 transformer on a custom-curated Telugu narrative dataset to generate coherent abstractive summaries, addressing linguistic challenges such as free-word order and multi-sentence compression.
- Evaluated model quality using ROUGE-1/2/L metrics and implemented preprocessing pipelines for tokenization, text normalization, and dataset balancing to improve summary fluency and content retention.

ACHIEVEMENTS AND ACTIVITIES

- Received Outstanding Academic Achievement Award during Masters in Computer Science at George Mason University.
- Won 1st place in the CSE Department Webathon technical competition during my Bachelors.